CURRENCY DETECTOR

Three-Page Evaluator Setup & Run Guide

1. Project Overview

This project uses a trained YOLOv8 object-detection model exported to ONNX format. The test application loads the ONNX file through OpenCV, reads live video from a webcam, and displays a detection box and confidence score when it identifies a ₹50 note.

The submitted test program is test.py. The ONNX model is stored at models/onnx/model.onnx in the project folder.

2. Original dataset and source verification

IMPORTANT: THE COMPLETE ORIGINAL DATASET IS AVAILABLE ON GOOGLE DRIVE. IT IS NOT INCLUDED IN THE GITHUB REPOSITORY BECAUSE OF ITS LARGE FILE SIZE.

Google Drive dataset link: REPLACE THIS LINE WITH THE ACTUAL GOOGLE DRIVE SHARING LINK

The Google Drive folder contains the original image collection used to prepare the project dataset. The evaluator is requested to inspect these images to review the samples and verify their source and collection details. The folder includes ₹50-note images (front, back and different conditions), negative samples of other denominations, and non-currency/background images.

The dataset is provided for transparency and evaluation. Please describe the image sources accurately: identify which images were personally captured and which, if any, were obtained from online sources. Do not represent online images as personally captured.

The dataset is needed to inspect or reproduce training, but it is not required for normal webcam inference. The evaluator can run test.py using the submitted ONNX model without downloading the dataset.

2. Software and technologies used

Technology / tool	Purpose in this project
Python	Programming language used for training utilities and the webcam test program.

YOLOv8n (Ultralytics)	Lightweight object-detection model trained to identify the ₹50-note class.
OpenCV (cv2.dnn)	Loads the ONNX model and performs webcam image inference.
NumPy	Supports image and model-output array processing.
ONNX	Portable model format used to export and submit the trained model.
LabelImg + YOLO labels	Used to draw note bounding boxes and prepare training annotations.

3. What was completed

- Collected and labelled positive ₹50-note images and negative images such as other denominations and non-currency objects.
- Prepared a dataset of 574 images, split into 401 training, 114 validation and 59 test images.
- Trained a YOLOv8n model using 640 × 640 input images for up to 100 epochs.
- Evaluated the trained model. The recorded held-out test results were: precision 99.8%, recall 100.0%, mAP@50 99.5%, and mAP@50–95 83.7%.
- Exported the trained weights to ONNX format (opset 12) and saved the model as models/onnx/model.onnx.
- Created test.py to load the ONNX model with OpenCV and run live webcam detection.

5. Important evaluation note

The current training class is named 50_rupee and was trained on general ₹50-note examples. The metrics above measure that class on the prepared test split; they do not prove that the model rejects every older ₹50 design while accepting only the new light blue-green design. That specific old-versus-new distinction must be verified with representative old and new note images.

6. Download the project from GitHub

Before submission, replace the placeholder below with the actual public GitHub repository URL and make sure the repository contains test.py and models/onnx/model.onnx.

```
https://github.com/YOUR-GITHUB-USERNAME/YOUR-REPOSITORY
```

Option A — Download ZIP (no Git installation required):

1. Open the project repository page in a web browser.
2. Select the green Code button, then choose Download ZIP.
3. Extract the ZIP file to a folder on the computer.
4. Open PowerShell or Command Prompt in the extracted project folder.

Option B — Clone using Git:

```
git clone https://github.com/YOUR-GITHUB-USERNAME/YOUR-REPOSITORY.git
cd YOUR-REPOSITORY
```

7. Python requirements

Use Python 3.10–3.13 where possible. The evaluator needs Python, pip, OpenCV and NumPy to run the webcam test program. Model-training packages are not required for inference.

Check that Python and pip are available:

```
python --version
python -m pip --version
```

Install the runtime packages:

```
python -m pip install opencv-python numpy
```

If the project includes a requirements.txt file, the evaluator can instead install the listed dependencies with the following command:

```
python -m pip install -r requirements.txt
```

A working webcam must be connected and available to the computer. Close other applications that may already be using the webcam.

8. Check the project files

The important files should be arranged like this:

```
YOUR-REPOSITORY/\n ├── test.py\n └── models/\n     └── onnx/\n         └── model.onnx
```

The model file is required. If model.onnx is missing, test.py cannot perform detection. Do not rename or move it unless you also update the model path in the run command.

9. Run the webcam test

Open PowerShell or Command Prompt in the main project folder — the same folder that contains test.py.

Run using the model's current folder location:

```
python test.py --model models/onnx/model.onnx
```

The assignment also specifies the shorter command below. To use it, first copy the ONNX model to the project root:

```
Copy-Item .\models\onnx\model.onnx .\model.onnx
python test.py --model model.onnx
```

On successful startup, the webcam window should appear. Hold a note in front of the camera, keep the full note visible, and allow the camera to focus.

10. How to operate the test window

Key / display	What it means
Green bounding box	The region the model has identified as a ₹50 note.
50_rupee + confidence	The predicted class and model confidence score.
Q	Quit the webcam application.
S	Save a screenshot, if screenshot saving is enabled in the supplied test.py.

11. Suggested evaluation checks

Please test under more than one condition:

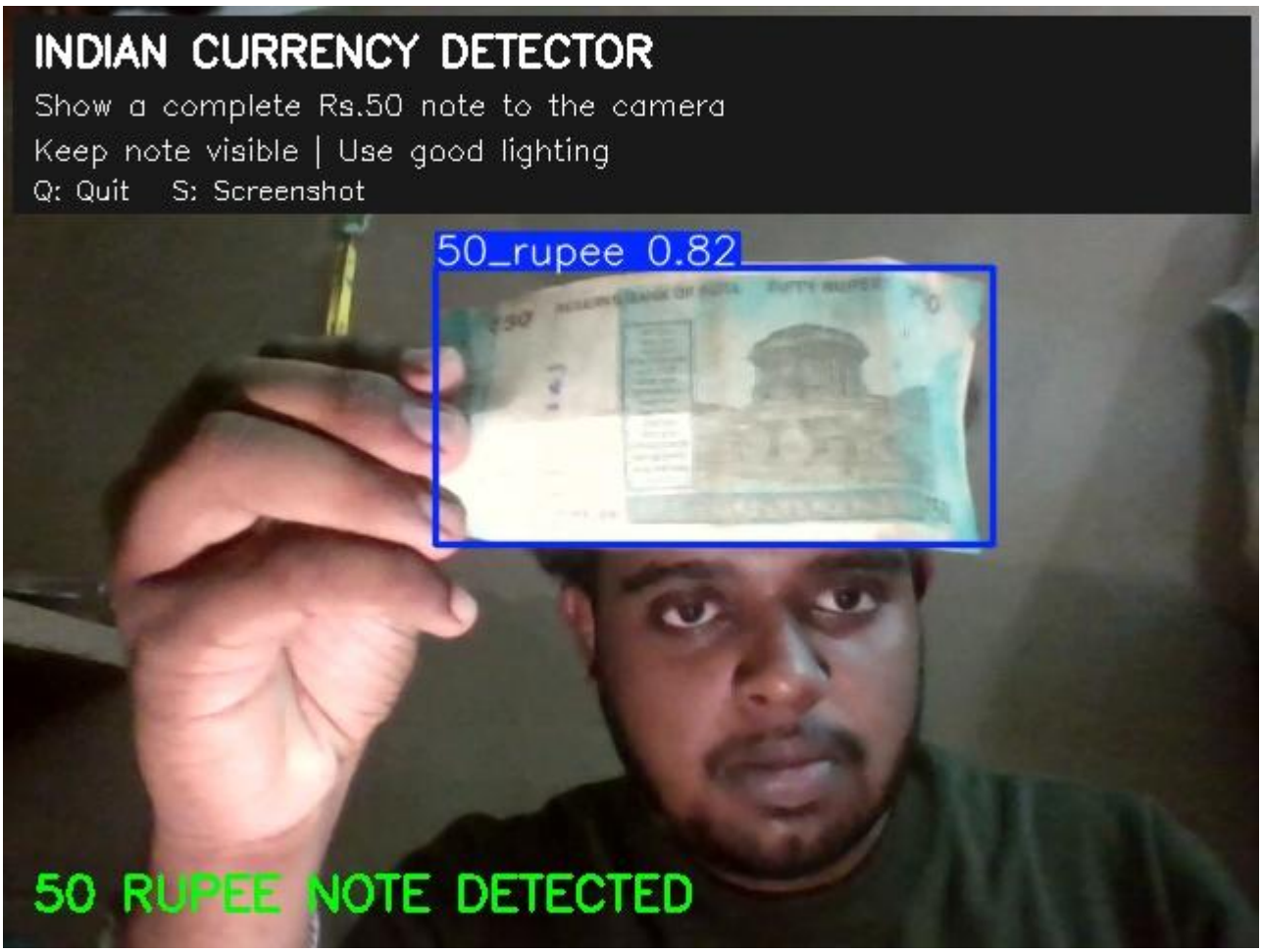
- Show the front and back of a ₹50 note, with the note fully visible.
- Move the note slightly farther away and closer to the webcam.
- Try different lighting conditions and backgrounds.
- Show other denominations (₹10, ₹20, ₹100, ₹200 and ₹500) and ordinary objects.
- If the requirement is to accept only the new ₹50 design, separately test older ₹50 notes and record whether they are rejected.

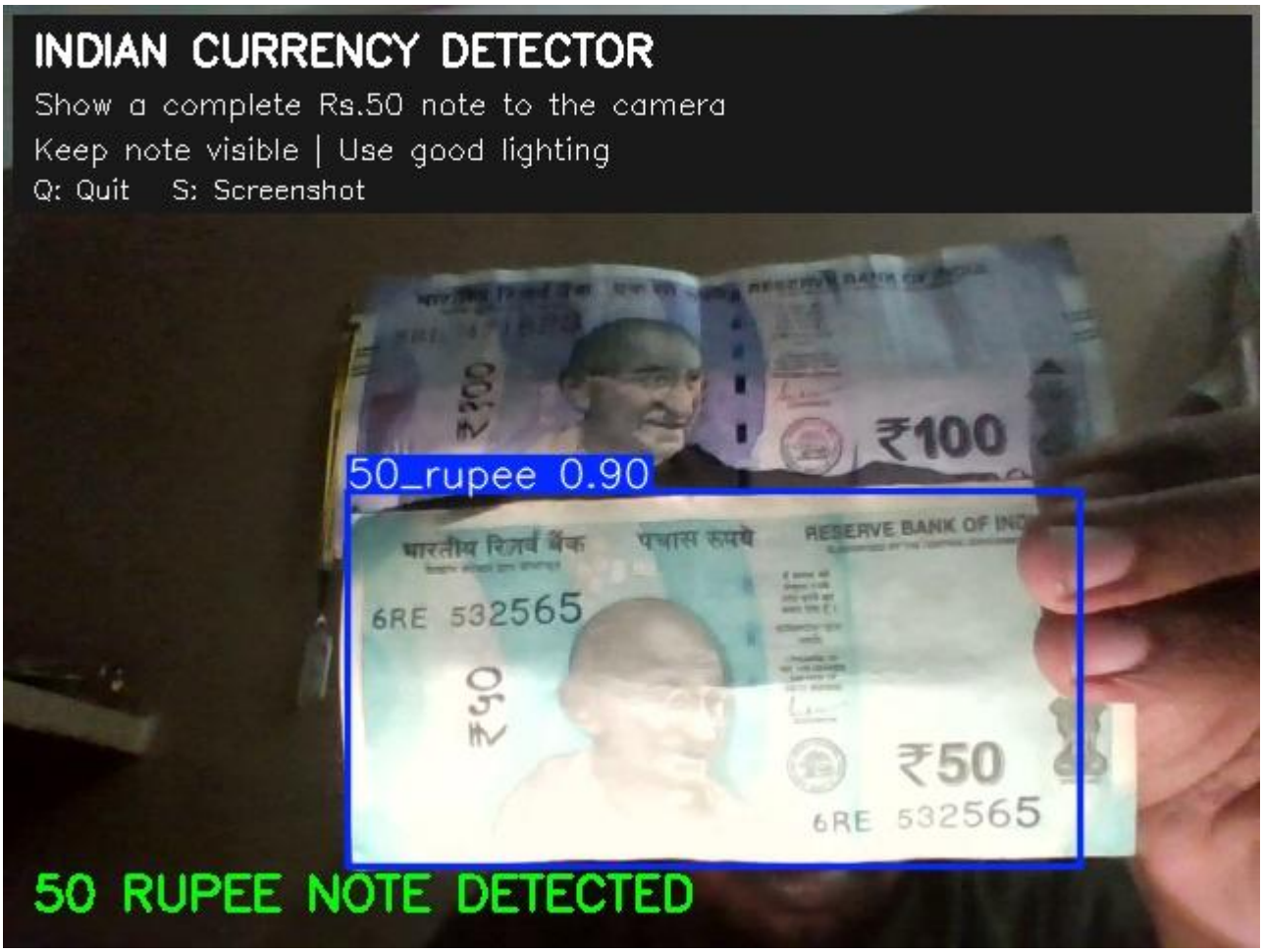
12. Troubleshooting

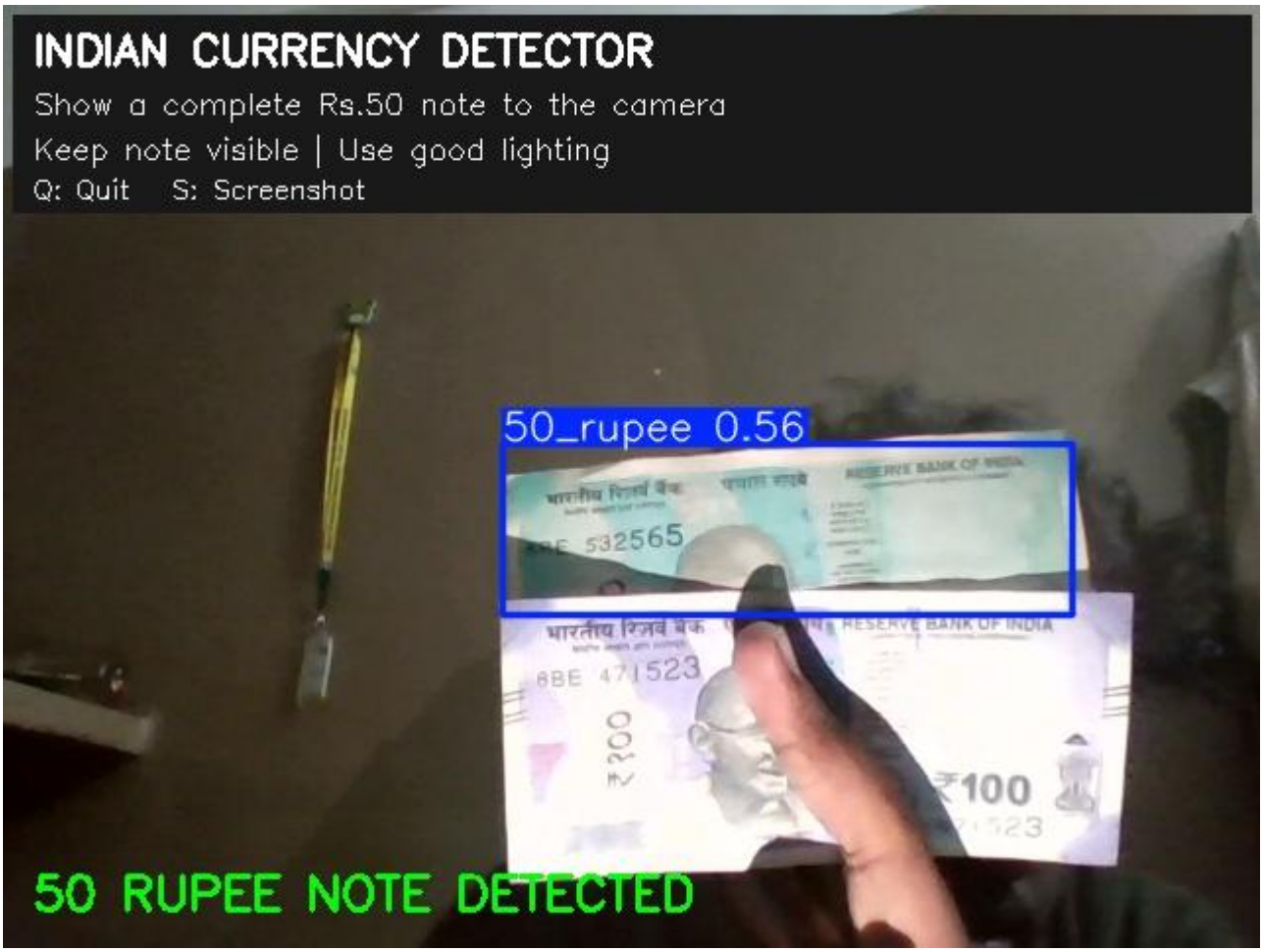
- “No module named cv2”: run python -m pip install opencv-python.
- “No module named numpy”: run python -m pip install numpy.
- Model file not found: check the current folder and the path after --model.
- Camera does not open: close other camera applications, reconnect the webcam, and try again.
- Poor or inconsistent detections: improve lighting, show the whole note, avoid motion blur, and test with more than one sample.

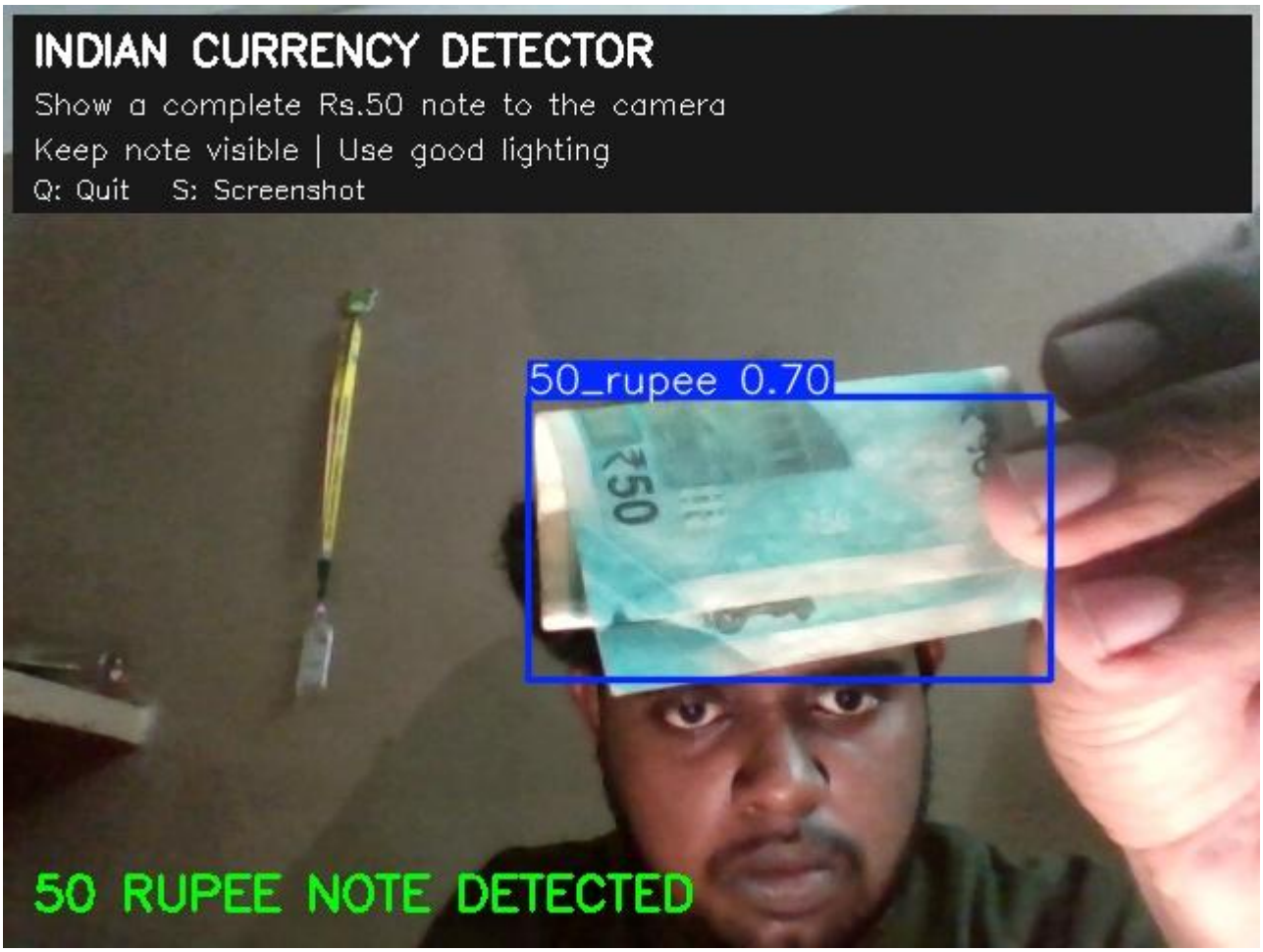
Submission files: test.py and the exported ONNX model (model.onnx). Keep the model path and run command consistent.

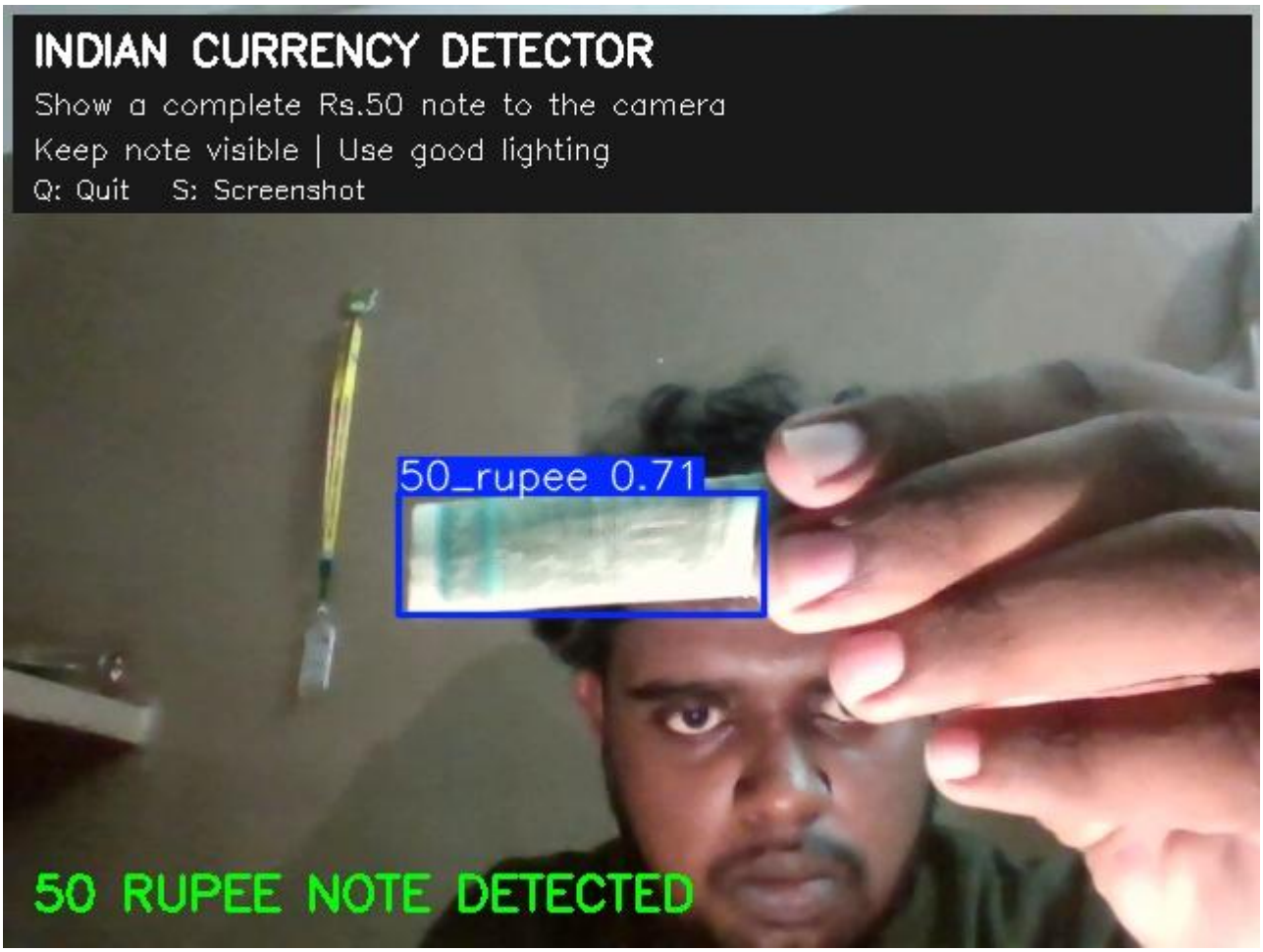
12. SAMPLE DEMO IMAGES

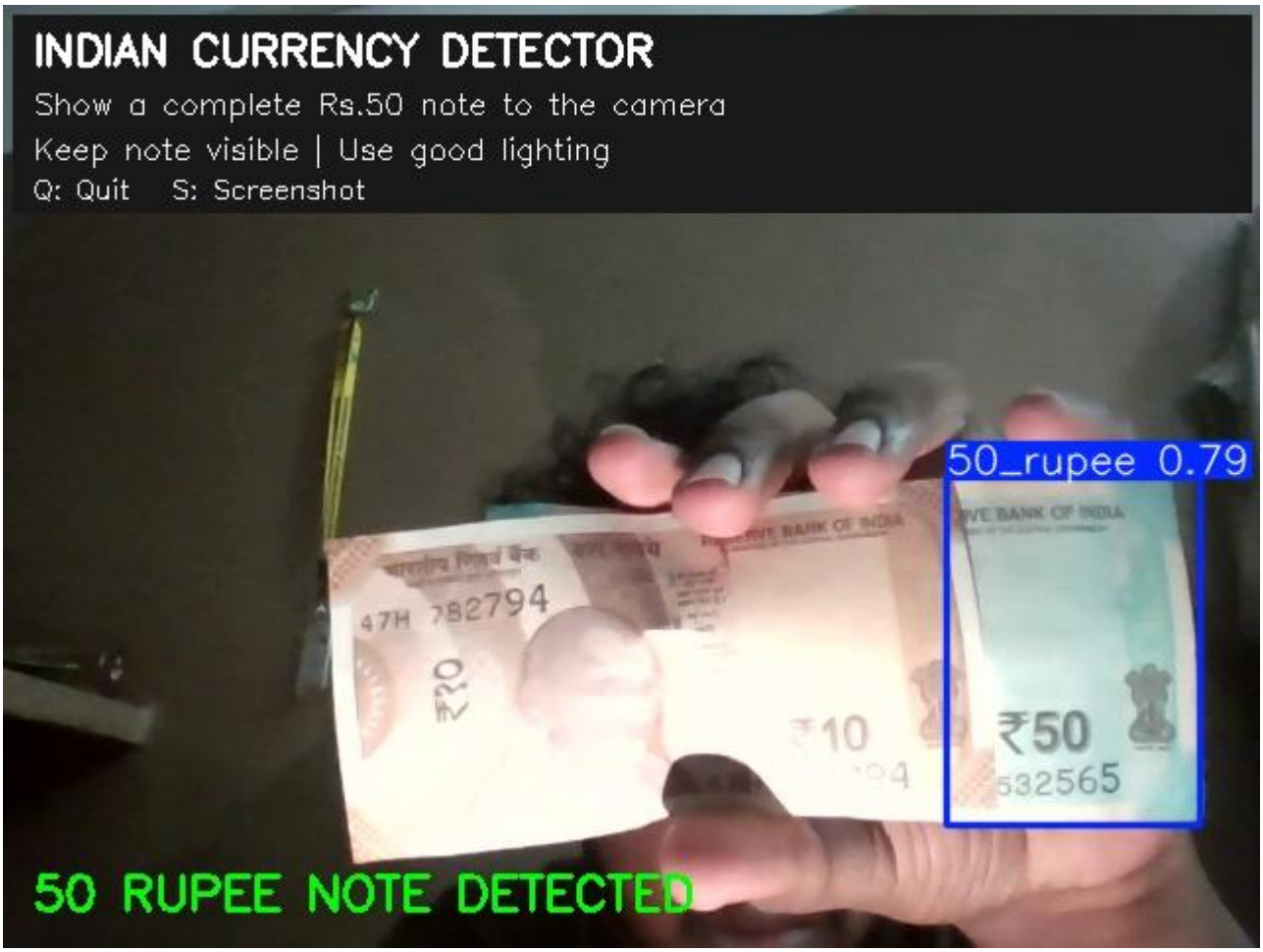



INDIAN CURRENCY DETECTOR
Show a complete Rs.50 note to the camera
Keep note visible | Use good lighting
Q: Quit S: Screenshot
50_rupee 0.90
₹100
₹50
6RE 532565
50 RUPEE NOTE DETECTED


INDIAN CURRENCY DETECTOR
Show a complete Rs.50 note to the camera
Keep note visible | Use good lighting
Q: Quit S: Screenshot
50_rupee 0.56
532565
8BE 471523
₹100
50 RUPEE NOTE DETECTED


INDIAN CURRENCY DETECTOR
Show a complete Rs.50 note to the camera
Keep note visible | Use good lighting
Q: Quit S: Screenshot
50_rupee 0.70
50 RUPEE NOTE DETECTED


INDIAN CURRENCY DETECTOR
Show a complete Rs.50 note to the camera
Keep note visible | Use good lighting
Q: Quit S: Screenshot
50_rupee 0.71
50 RUPEE NOTE DETECTED


INDIAN CURRENCY DETECTOR
Show a complete Rs.50 note to the camera
Keep note visible | Use good lighting
Q: Quit S: Screenshot
50_rupee 0.79
50 RUPEE NOTE DETECTED


INDIAN CURRENCY DETECTOR
Show a complete Rs.50 note to the camera
Keep note visible | Use good lighting
Q: Quit S: Screenshot
50_rupee 0.80
50 RUPEE NOTE DETECTED